EXPERIMENT 12

Non-Contiguous Memory Allocation: OPTIMAL

AIM:

To implement the Optimal Page Replacement algorithm in C and calculate the total number of page faults.

PROCEDURE:

1.Start the process

2.Declare the size

3.Get the number of pages to be inserted

4.Get the value

5.Declare counter and stack

6.Select the least frequently used page by counter value

7.Stack them according the selection.

8.Display the values

9.Stop the process

PROGRAM:

```
#include <stdio.h>
int main()
{
    int frames[10], pages[30];
    int n, m, i, j, k, pos, faults = 0;
    int found, farthest, next;

    printf("Enter number of frames: ");
    scanf("%d", &m);

    printf("Enter number of pages: ");
    scanf("%d", &n);

    if(m < 1 || m > 10 || n < 1 || n > 30)
    {
        printf("Invalid input size\n");
        return 1;
    }

    printf("Enter reference string: ");
```

```
for(i = 0; i < n; i++)
    scanf("%d", &pages[i]);

for(i = 0; i < m; i++)
    frames[i] = -1;

for(i = 0; i < n; i++)
{
    found = 0;

    for(j = 0; j < m; j++)
    {
        if(frames[j] == pages[i])
        {
            found = 1;
            break;
        }
    }

    if(found == 0)
    {
        faults++;
        pos = -1;

        for(j = 0; j < m; j++)
        {
            if(frames[j] == -1)
            {
                pos = j;
                break;
            }
        }

        if(pos == -1)
        {
            farthest = -1;

            for(j = 0; j < m; j++)
            {
                for(k = i + 1; k < n; k++)
                {
                    if(frames[j] == pages[k])
                        break;
                }

                next = k;

                if(next > farthest)
                {
                    farthest = next;
                    pos = j;
```

```
                    }
                }
            }

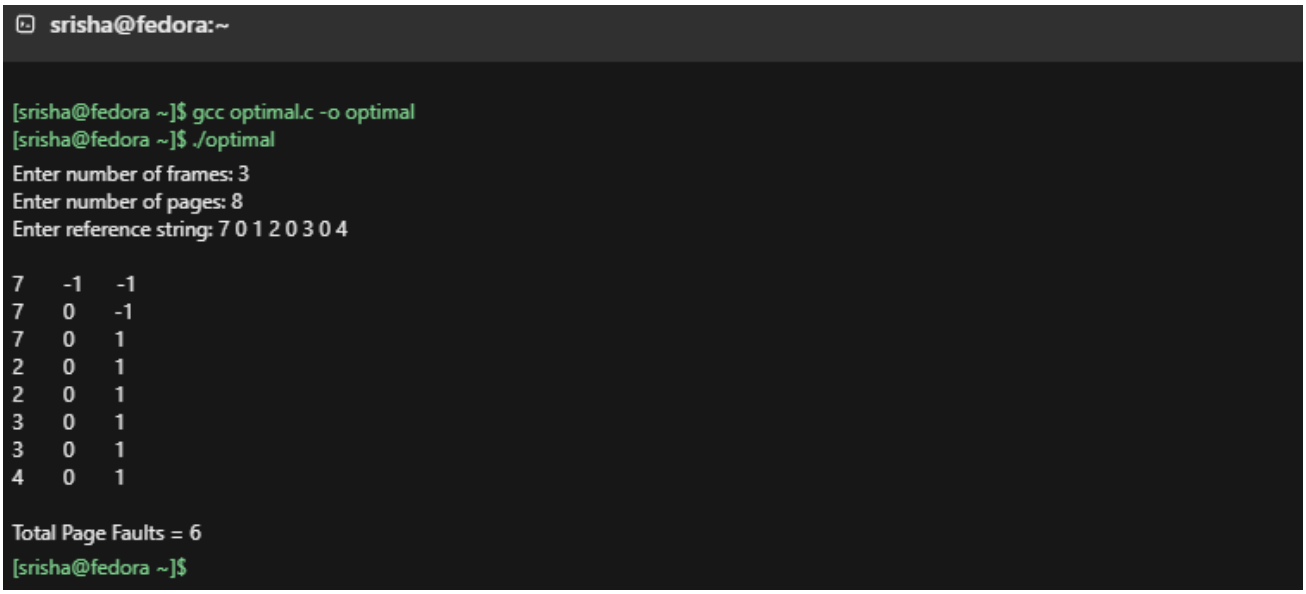
            frames[pos] = pages[i];
        }

        printf("\n");
        for(j = 0; j < m; j++)
            printf("%d\t", frames[j]);
    }

    printf("\n\nTotal Page Faults = %d\n", faults);

    return 0;
}
```

OUTPUT:



RESULT:

Thus, the Optimal Page Replacement algorithm was implemented and executed successfully using C in Fedora Linux, and the total number of page faults was calculated.